ProTutor

A Project Report submitted in partial fulfilment of the requirements for the award of the degree of

Master of Computer Applications

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Declaration

I hereby declare that the work which is being presented in the M.C.A. Project **"Protutor"**, in partial fulfilment of the requirements for the award of the ***Master of Computer Applications*** and submitted to the Department of Computer Engineering and Applications of GLA University, Mathura, is an authentic record of my own work carried under the supervision of **Mr. Nitish Bhardwaj (Assistant Professor).**

The contents of this project report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree.

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Abstract

ProTutor is a full-stack, AI-powered learning platform designed to provide students with a centralized system for improving their academic performance. In today's digital environment, students often rely on multiple platforms for studying, practicing, and solving doubts. This scattered approach leads to inefficiency, time loss, and lack of proper guidance. ProTutor addresses this issue by integrating all essential learning tools into a single, user-friendly platform.

The system is developed using the PERN stack, which includes PostgreSQL, Express.js, React.js, and Node.js. This ensures a scalable, reliable, and high-performance application. The frontend is built using React.js and styled with Tailwind CSS to provide a clean, responsive, and modern user interface. The backend manages user authentication, data processing, and communication with the database effectively.

ProTutor offers several intelligent features powered by artificial intelligence. These include AI-based doubt solving, where students can ask questions and receive instant answers; automated quiz generation for practice; smart content creation for study materials; and performance analytics to track progress and identify weak areas. The platform also includes resume guidance to support students in their career preparation.

From a business perspective, ProTutor follows a Software as a Service (SaaS) model. Users can register securely using email or social login and access a personalized dashboard. Free users have limited access, while premium users can unlock advanced features such as unlimited queries, detailed analytics, and enhanced learning tools through a subscription or credit-based system.

Overall, ProTutor represents a complete full-stack development project that combines frontend, backend, database, and AI integration. It provides a practical and efficient solution for students while demonstrating real-world software development skills.

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CHAPTER 1
Introduction

1.1 Overview

In today's digital world, technology has become an important part of education. Students now depend on online platforms for learning, practicing questions, and clearing doubts. However, most students use multiple applications such as YouTube for lectures, Google for doubts, quiz apps for practice, and note-making tools for revision. This creates confusion and makes learning less efficient.

ProTutor is a full-stack AI-powered learning platform that solves this problem by providing all these features in one place. It is designed to help students learn in a smarter and more organized way. The platform provides a centralized dashboard where students can access study materials, solve doubts, generate quizzes, track performance, and even get resume guidance.

The system is developed using the PERN stack, which includes PostgreSQL (database), Express.js and Node.js (backend), and React.js (frontend). This technology ensures that the application is fast, scalable, and user-friendly. The user interface is designed using Tailwind CSS, which makes it responsive and easy to use on different devices.

ProTutor also uses Artificial Intelligence (AI) to enhance learning. AI helps in solving doubts instantly, generating questions, creating summaries, and analyzing performance. This makes the learning experience more personalized and effective.

Overall, ProTutor is a modern solution that combines technology and education to provide a better learning experience for students.

1.2 Motivation

The main motivation behind developing ProTutor is to solve the problems faced by students during online learning.

Students often face the following issues:

- They have to switch between multiple apps for different purposes.

- There is no personalized guidance for learning.
- They cannot track their performance properly.
- They spend too much time searching for study material.
- There is no single platform that provides all features together.

These problems make learning difficult and time-consuming. Students lose focus and become less productive.

ProTutor is created to overcome these challenges. The idea is to build a single platform that provides all learning tools in one place. With AI integration, the platform can understand student needs and provide better support.

Another motivation is to create a real-world full-stack project. This project helps in understanding how modern web applications are built, including frontend development, backend logic, database management, authentication, and AI integration.

Thus, ProTutor is not only useful for students but also serves as a practical learning experience in software development.

1.2 Problem Statement

Despite the availability of many online learning platforms, there are still several problems that students face.

The major problems are:

1. **Scattered Learning Resources**
   Students use different platforms for videos, notes, quizzes, and doubt solving. This makes learning unorganized.
2. **Lack of Personalization**
   Most platforms provide the same content to all users. They do not adapt to individual learning needs.
3. **Time Consumption**
   Switching between multiple platforms wastes time and reduces productivity.

4. **No Centralized Dashboard**
   There is no single system where students can track all their activities and progress.
5. **Limited AI Support**
   Many platforms do not use AI effectively to help students.
6. **Poor Performance Tracking**
   Students cannot properly analyze their strengths and weaknesses.

Because of these problems, students are not able to learn efficiently. There is a need for a system that can provide all features in one place and improve the overall learning experience.

1.3 Objectives

The main objective of the ProTutor project is to develop a smart and efficient learning platform.

The specific objectives are:

- To develop a centralized platform for learning
- To integrate AI-based tools for doubt solving and content generation
- To provide automated quiz generation for practice
- To track student performance using analytics
- To reduce time spent on switching platforms
- To provide a simple and user-friendly interface
- To implement a secure authentication system
- To introduce a subscription-based model for premium features

The system aims to make learning faster, easier, and more effective for students.

1.4 Scope of the Project

The scope of ProTutor defines what the system will do and what it will not do.

In-Scope

- Web-based application
- User registration and login system
- AI-based doubt solving
- Quiz generation system
- Performance tracking and analytics
- Content generation for study materials
- Resume guidance
- Subscription and credit-based system

Out-Scope

- Mobile application (future scope)
- Live video classes
- Offline functionality
- Advanced AI model training

The project focuses mainly on building a web platform with essential features.

1.5 Advantages of the System

ProTutor provides several advantages:

- **Centralized Platform** – All tools are available in one place
- **Time Saving** – No need to switch between apps
- **AI Support** – Instant answers and smart suggestions
- **Better Learning Experience** – Personalized features
- **Performance Tracking** – Helps students improve
- **User-Friendly Interface** – Easy to use for all students

These advantages make ProTutor more efficient compared to traditional learning methods.

1.6 Organization of the Project Report

This project report is organized into different chapters to explain the system clearly and in a structured manner. Each chapter focuses on a specific part of the project so that the reader can easily understand the development process step by step.

Chapter 1 explains the introduction, motivation, problem statement, objectives, and scope of the project. It provides a basic understanding of why the project was developed and what problems it aims to solve. This chapter also highlights the importance of the system and gives an overview of its features. It helps the reader understand the purpose and need of the project. Additionally, it introduces the technologies used and the overall concept of the system.

Chapter 2 describes the literature review and comparison with existing systems. It explains different platforms or methods currently used for learning and identifies their limitations. This chapter helps in understanding the gap in existing solutions. It also justifies the need for the proposed system. A comparison is made to show how ProTutor improves upon existing systems. This chapter builds a strong foundation for the project.

Chapter 3 explains system requirements and analysis in detail. It includes functional requirements, non-functional requirements, and feasibility analysis. This chapter defines what the system will do and how it will perform. It also includes hardware and software requirements needed for development. Risk analysis is also discussed to identify possible issues. This ensures proper planning before development begins.

Chapter 4 describes the system design including diagrams. It explains how the system is structured and how different components interact with each other. Diagrams such as DFD, UML, and ER are included to represent the system visually. This chapter helps in understanding the internal working of the system. It also defines modules and their responsibilities. Proper design ensures smooth implementation.

Chapter 5 explains the implementation and AI features of the system. It describes how the frontend, backend, and database are developed. This chapter also explains how AI features like doubt solving, quiz generation, and analytics are implemented. It provides

details about authentication and subscription systems. The working of each module is clearly explained. This chapter shows how the design is converted into a real system.

Chapter 6 covers testing and validation of the system. It explains different types of testing such as unit testing and integration testing. This chapter ensures that all features are working correctly. Test cases are used to verify system functionality. It also helps in identifying and fixing errors. Proper testing improves system reliability and performance.

Chapter 7 concludes the project. It summarizes the overall work done in the project. This chapter highlights the achievements and benefits of the system. It also suggests possible improvements and future enhancements. The scope for adding new features is discussed. This chapter provides a final understanding of the project.

CHAPTER 2
Literature Review

2.1 Existing Systems

In the current digital era, many online learning platforms are available that help students in their studies. These platforms provide different types of services such as video lectures, practice questions, doubt solving, and study materials. Some popular examples include educational websites, mobile learning apps, and online course platforms. These systems have made learning more accessible and flexible for students.

Most existing systems focus on a specific type of learning. For example, some platforms provide only video lectures, while others focus only on quizzes or mock tests. Similarly, some applications are used only for doubt solving, where students can ask questions and get answers. While these platforms are useful, they do not provide all features in one place.

Another important aspect of existing systems is that they are not fully personalized. Most platforms provide the same content to all users, without considering individual learning needs. This means that students with different levels of understanding receive the same type of content, which may not be effective for everyone.

Many platforms also lack proper performance tracking. Although some systems provide basic results, they do not offer detailed analysis of student performance. Students are not able to identify their weak areas or track their progress over time effectively.

In addition, existing systems often require students to switch between multiple platforms. For example, a student may watch lectures on one app, solve quizzes on another, and ask doubts on a different platform. This makes the learning process unorganized and time-consuming.

Overall, while existing systems provide useful features, they do not offer a complete and integrated solution for students. This creates a need for a platform that combines all essential learning tools into one system.

Table 2.1: Comparison of Existing Tutoring Platform with ProTutor

Feature	Existing Platform	Proposed System (ProTutor)
User Registration	Available	Available
Tutor Search	Basic Search	Advanced Search
Session Booking	Limited Options	Flexible & Easy Booking
Online Payment	Available	Secure & Fast Payment
Dashboard	Limited	Advanced Dashboard
AI Chat Support	Not Available	Available
Progress Tracking	Basic	Detailed Tracking
User Interface	Average	User-Friendly

2.2 Limitations of Existing Systems

Despite the availability of many online learning platforms, there are several limitations that affect their effectiveness. These limitations make it difficult for students to achieve the best learning outcomes.

One of the major limitations is the lack of integration. Most platforms focus on a single feature such as videos, quizzes, or notes. Students have to use different platforms for different tasks, which leads to confusion and inefficiency. There is no centralized system that brings all features together.

Another limitation is the lack of personalization. Existing systems do not adapt to the individual needs of students. Every student has a different learning style, pace, and level of understanding. However, most platforms provide the same content to all users. This reduces the effectiveness of learning.

Time consumption is also a major issue. Students spend a lot of time switching between applications, searching for content, and managing different tools. This reduces productivity and affects focus.

Limited use of Artificial Intelligence is another drawback. Although some platforms use AI, it is often limited to basic recommendations. Advanced features such as smart

doubt solving, automatic content generation, and detailed analytics are not widely available.

Performance tracking in existing systems is also not very effective. Students are not able to get detailed insights into their progress. They cannot easily identify their strengths and weaknesses. This makes it difficult to improve performance.

Security and authentication can also be a concern in some platforms. Not all systems provide secure login methods or protect user data properly. This can lead to data privacy issues.

Finally, many platforms do not provide career-related support such as resume building or guidance. Students need to use separate tools for career preparation, which again increases dependency on multiple systems.

These limitations clearly show that there is a need for a better system that can overcome these challenges and provide a complete learning solution.

2.3 Proposed System (ProTutor)

To overcome the limitations of existing systems, the ProTutor platform is proposed as a complete and integrated solution for students. ProTutor is a full-stack AI-powered learning platform that combines all essential academic tools into one system.

The main goal of ProTutor is to provide a centralized platform where students can perform all learning activities without switching between multiple applications. The system includes features such as AI-based doubt solving, quiz generation, content creation, performance tracking, and resume guidance.

One of the key features of ProTutor is the use of Artificial Intelligence. The AI system helps students by providing instant answers to their questions. It also generates quizzes based on selected topics, helping students practice effectively. In addition, AI can create summaries and notes, making it easier for students to revise.

Another important feature is personalized learning. ProTutor tracks user activity and performance to provide better suggestions and insights. Students can identify their weak areas and focus on improving them. This makes learning more efficient and targeted.

The platform also includes a performance analytics system. This system provides detailed reports on student progress. It helps students understand their strengths and weaknesses and improve their performance over time.

ProTutor is built using the PERN stack, which ensures high performance and scalability. The frontend is developed using React.js with Tailwind CSS for a modern and responsive user interface. The backend uses Node.js and Express.js to handle logic and API requests, while PostgreSQL is used for database management.

The system also includes a secure authentication mechanism. Users can register and log in using email or social accounts such as Google. This ensures data security and ease of access.

From a business perspective, ProTutor follows a SaaS-based model. Users can access basic features for free, while premium features are available through subscriptions or a credit-based system. This makes the platform flexible and accessible to a wide range of users.

In addition, ProTutor includes resume guidance features to help students prepare for their careers. This makes it more than just a learning platform, as it also supports professional development.

Overall, the proposed system provides a complete solution that addresses all the limitations of existing platforms. It offers a unified, efficient, and intelligent learning environment for students.

CHAPTER 3
Software Requirement And Analysis

3.1 Functional Requirements

Functional requirements define what the system should do. These are the core features of the ProTutor platform that allow users to interact with the system and perform different activities.

The first important requirement is **user registration and login**. The system must allow users to create an account using email or social login such as Google. After registration, users should be able to log in securely and access their dashboard.

The second requirement is the **user dashboard**. After login, users should see a personalized dashboard that shows all features such as quizzes, AI tools, analytics, and study materials. The dashboard should be simple and easy to use.

Another important feature is **AI-based doubt solving**. Users should be able to ask questions and get instant answers using AI. This feature helps students understand concepts quickly without searching multiple sources.

The system must also include a **quiz generation feature**. Users should be able to generate quizzes based on topics or subjects. The system should automatically create questions and provide results after completion.

Performance tracking is another key requirement. The system should store user activity and provide analytics such as scores, progress, and weak areas. This helps students improve their performance.

The platform should also provide **content generation** features. Users can generate summaries, notes, or explanations for study topics using AI.

Another requirement is the **subscription system**. The platform should support both free and premium users. Free users get limited access, while premium users can unlock advanced features.

Finally, the system should include **resume guidance** features. Users should be able to create or improve their resumes using the platform.

These functional requirements ensure that the system provides all necessary features for learning and development.

Table 3.1: Functional Requirements

S. No	Function	Description
1	User Registration	User can create a new account
2	User Login	User can login using email and password
3	Tutor Search	User can search tutors based on subjects
4	View Tutor Details	User can view tutor profile and details
5	Session Booking	User can book a tutoring session
6	Online Payment	User can make payment for session
7	Dashboard View	User can view performance and session details
8	AI Chat Support	User can interact with chatbot for help

3.2 Non-Functional Requirements

Non-functional requirements define how the system should perform. These requirements focus on quality, performance, and user experience.

One of the most important non-functional requirements is **performance**. The system should respond quickly to user actions. Pages should load fast, and AI responses should be generated without delay.

Another important requirement is **security**. User data such as login details and personal information must be protected. The system should use secure authentication methods and encryption to prevent unauthorized access.

Usability is also very important. The platform should be easy to use, even for beginners. The interface should be simple, clear, and user-friendly so that students can easily navigate through features.

Scalability is another requirement. The system should be able to handle a large number of users at the same time without performance issues. As the number of users grows, the system should still work efficiently.

Reliability is also important. The system should work properly without crashes or errors. It should provide consistent results and maintain uptime.

Compatibility ensures that the platform works on different devices and browsers. Users should be able to access the system from desktops, laptops, and mobile devices.

These non-functional requirements ensure that the system is efficient, secure, and user-friendly.

3.3 Feasibility Study

A feasibility study is done to check whether the project is practical and possible to implement. It includes technical, economic, and operational feasibility.

Technical Feasibility

The technologies used in this project, such as React.js, Node.js, and PostgreSQL, are widely used and well-supported. Developers have access to tools and resources to build the system. AI integration can also be implemented using available APIs. Therefore, the project is technically feasible.

Economic Feasibility

The cost of development is low because most tools and technologies used are open-source. There is no need for expensive software or hardware. The SaaS model also allows the system to generate revenue through subscriptions. Therefore, the project is economically feasible.

Operational Feasibility

The system is designed to be simple and user-friendly. Students can easily use the platform without special training. The features are easy to understand and operate. Therefore, the project is operationally feasible.

Overall, the feasibility study shows that the project can be successfully developed and implemented.

3.4 Hardware and Software Requirements

This section describes the hardware and software needed to develop and run the ProTutor system.

Hardware Requirements

Table 3.4.1: Hardware Requirements

S. No	Hardware Component	Specification
1	Processor	Intel i3 or above
2	RAM	4 GB or more
3	Hard Disk	500 GB or more
4	Monitor	15 inch or above
5	Keyboard	Standard Keyboard
6	Mouse	Standard Mouse
7	Internet Connection	Required for online access

Software Requirements

Table 3.4.2: Software Requirements

S. No	Software	Specification
1	Operating System	Windows 10 or above
2	Programming Language	Java / JavaScript
3	IDE	VS Code / IntelliJ IDEA
4	Database	MySQL / localStorage
5	Browser	Chrome / Edge
6	Version Control	Git (optional)
7	Tools	Node.js, npm

3.5 Risk Analysis

Risk analysis helps identify possible problems that may occur during development or usage of the system.

One major risk is **server failure**. If the server goes down, users will not be able to access the system. This can be solved by using backup servers and cloud hosting.

Another risk is **data security issues**. User data may be at risk if proper security measures are not taken. This can be prevented by using encryption and secure authentication methods.

AI errors are also a possible risk. AI may sometimes provide incorrect answers. This can be reduced by improving algorithms and validating responses.

System bugs can also affect performance. Proper testing should be done to identify and fix errors before deployment.

Another risk is **high user load**. If too many users access the system at the same time, it may slow down. This can be managed by optimizing performance and using scalable servers.

Finally, there is a risk of **user adoption**. Some users may find it difficult to switch to a new platform. This can be solved by providing a simple interface and good user experience.

By identifying these risks and planning solutions, the system can be made more reliable and efficient.

Figure 3.5: Risk Analysis

Sr. No	Risk	Impact	Solution
1	System Crash	Data loss Regular	backup
2	Network Failure	Service interruption	Stable internet connection
3	Unauthorized Access	Security breach	Use authentication
4	Payment Failure	Booking not completed	Retry payment option
5	Server Down	System unavailable	Use cloud backup

CHAPTER 4
System Design

4.1 System Architecture

System architecture defines the overall structure of the ProTutor platform and how different components interact with each other. The ProTutor system follows a **three-tier architecture**, which includes the frontend, backend, and database layers.

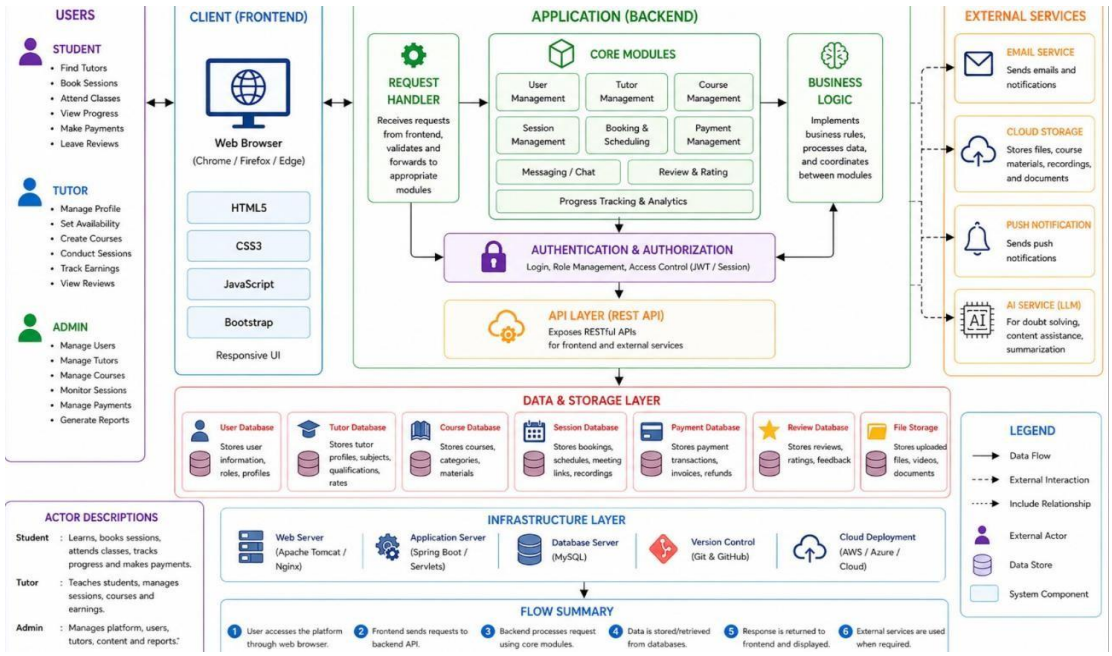


Figure 4.1: System architecture of Protutor

The **frontend layer** is the user interface of the system. It is developed using React.js and styled using Tailwind CSS. This layer is responsible for displaying information to the user and handling user interactions such as clicking buttons, submitting forms, and navigating between pages. The frontend communicates with the backend through API requests.

The **backend layer** is developed using Node.js and Express.js. It acts as the brain of the system and handles all the logic. It processes user requests, performs operations such as authentication, quiz generation, and AI processing, and sends responses back to the frontend. The backend also manages communication with the database.

The **database layer** uses PostgreSQL to store and manage data. It stores user information, quiz data, results, subscription details, and other necessary data. The database ensures that data is stored securely and can be accessed efficiently.

When a user performs an action, such as asking a question or starting a quiz, the request is sent from the frontend to the backend. The backend processes the request, interacts with the database if needed, and returns the result to the frontend. This architecture ensures smooth communication and efficient performance.

The three-tier architecture provides several advantages such as scalability, flexibility, and easy maintenance. Each layer can be updated independently without affecting the entire system.

4.2 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is used to represent how data moves through the system. It shows how information is processed, stored, and transferred between different components.

DFD Level 0 (Context Diagram)

The Level 0 DFD represents the system as a single process and shows its interaction with external entities. In ProTutor, the main external entity is the **User**.

- The user provides input such as login details, questions, and quiz requests.
- The system processes this input and provides output such as answers, quiz results, and analytics.

This level gives a high-level overview of the system.

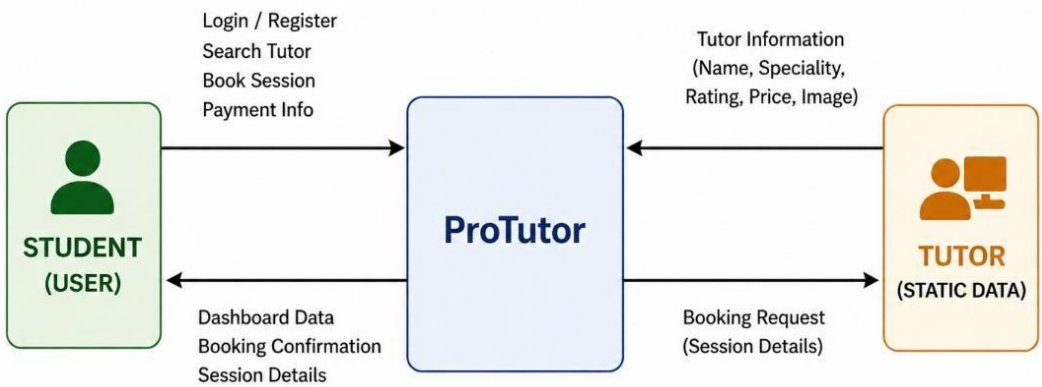


Figure 4.2.1: Data Flow Diagram (DFD - Level 0)

DFD Level 1

The Level 1 DFD provides a more detailed view of the system by breaking it into smaller processes.

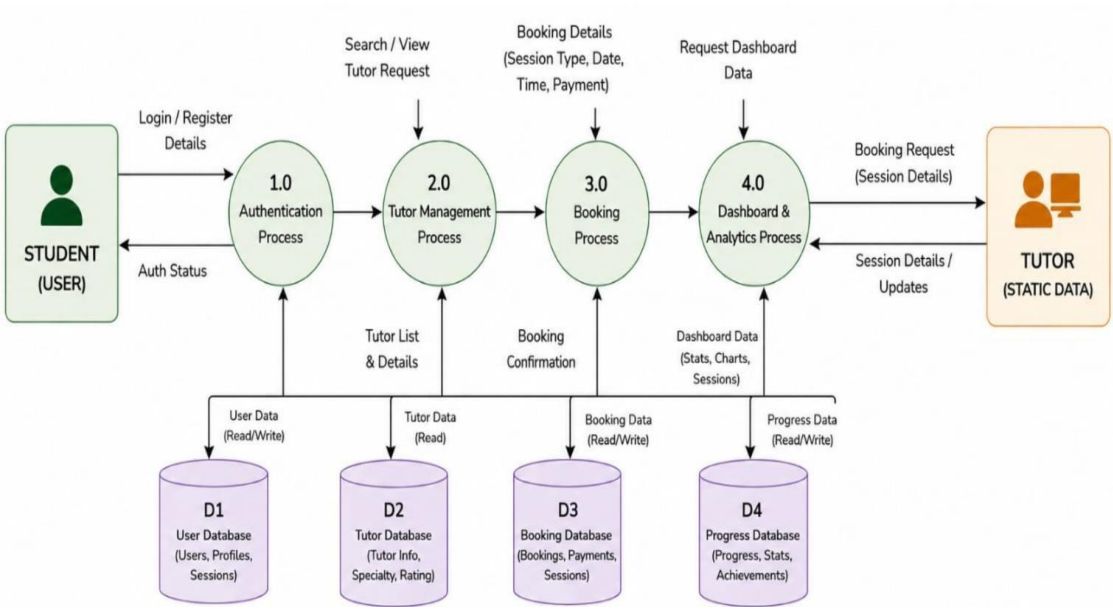


Figure 4.2.2: Data Flow Diagram (Level 1)

The main processes in ProTutor include:

1. **User Authentication Process**

   Handles registration and login. It verifies user credentials and allows access to the system.

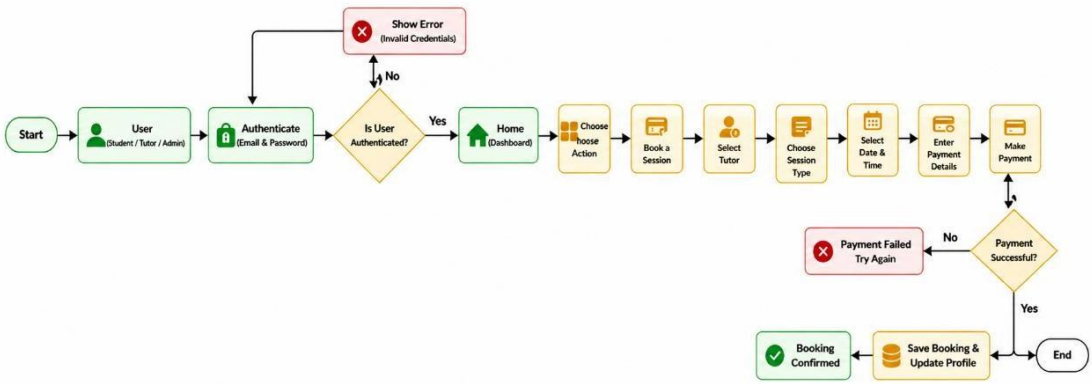


Figure 4.2.2.3: Activity Diagram - User Authentication

2. **AI Processing Module**

   Handles doubt solving and content generation. It takes user queries and returns AI-generated responses.

3. **Quiz Management Module**

   Generates quizzes, stores questions, and evaluates answers.

4. **Performance Tracking Module**

   Tracks user activity, stores results, and generates analytics.

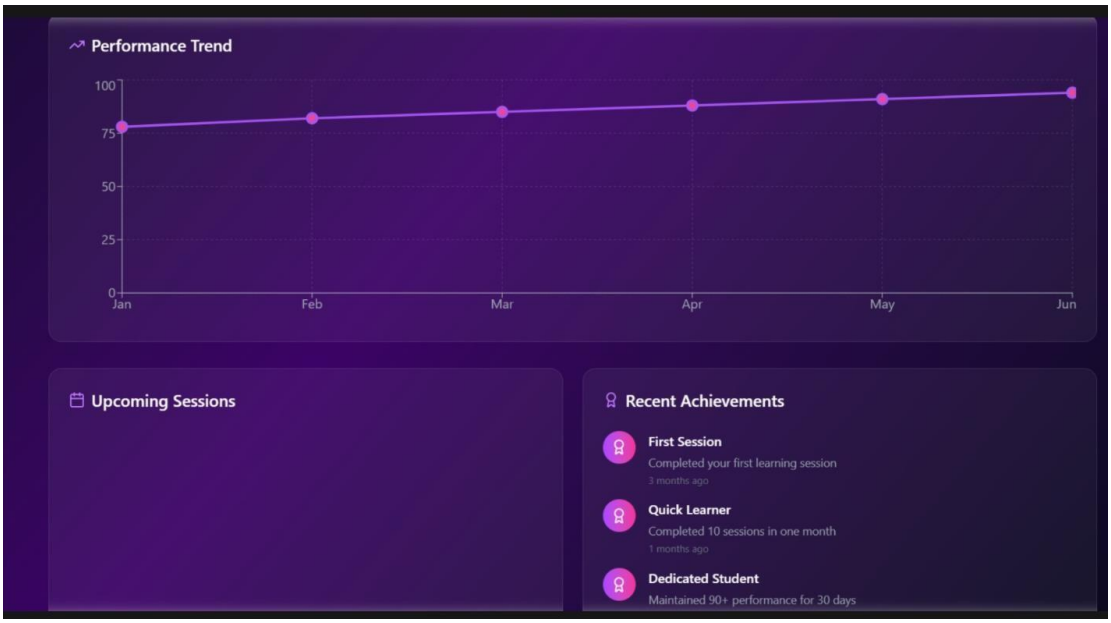


Figure 4.2.2.4: Performance Analytics Dashboard (Screenshot)

5. **Database Interaction**

   Stores and retrieves data such as user details, quiz results, and subscription information.

The DFD helps in understanding how data flows between these processes and ensures that the system is logically organized.

4.3 UML Diagram

UML (Unified Modeling Language) diagrams are used to represent the structure and behavior of the system. They help developers understand how different parts of the system are connected.

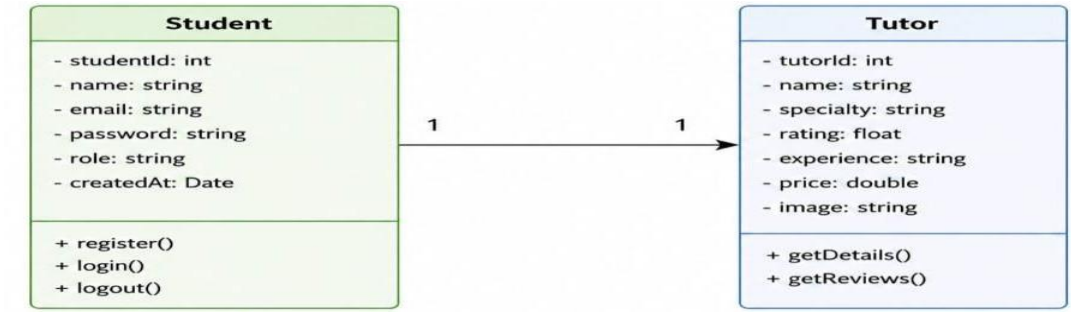


Figure 4.3.1: Unified Modeling Language (UML) Diagram

Use Case Diagram

The Use Case Diagram shows how users interact with the system. The main actor is the **User**. The user can perform actions such as:

- Register and login
- Ask questions (AI doubt solving)
- Generate quizzes
- View performance analytics
- Manage subscription

This diagram helps in identifying all functionalities of the system.

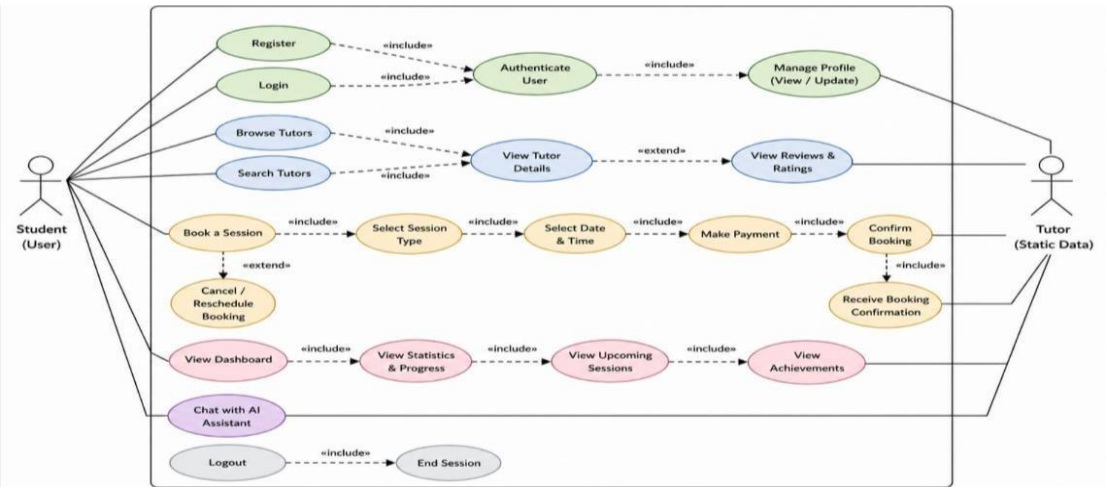


Figure 4.3.2

Class Diagram

The Class Diagram shows the structure of the system in terms of classes and their relationships. Some important classes include:

- **User Class** (stores user details)
- **Quiz Class** (stores quiz data)
- **Result Class** (stores performance data)
- **Subscription Class** (stores plan details)

These classes are connected with relationships such as association and dependency.

Sequence Diagram

The Sequence Diagram shows how processes interact over time. For example, in the login process:

1. User enters login details
2. Frontend sends request to backend
3. Backend verifies data with database
4. Response is sent back to user

This diagram helps in understanding the flow of operations step by step.

UML diagrams are important for visualizing system design and improving clarity.

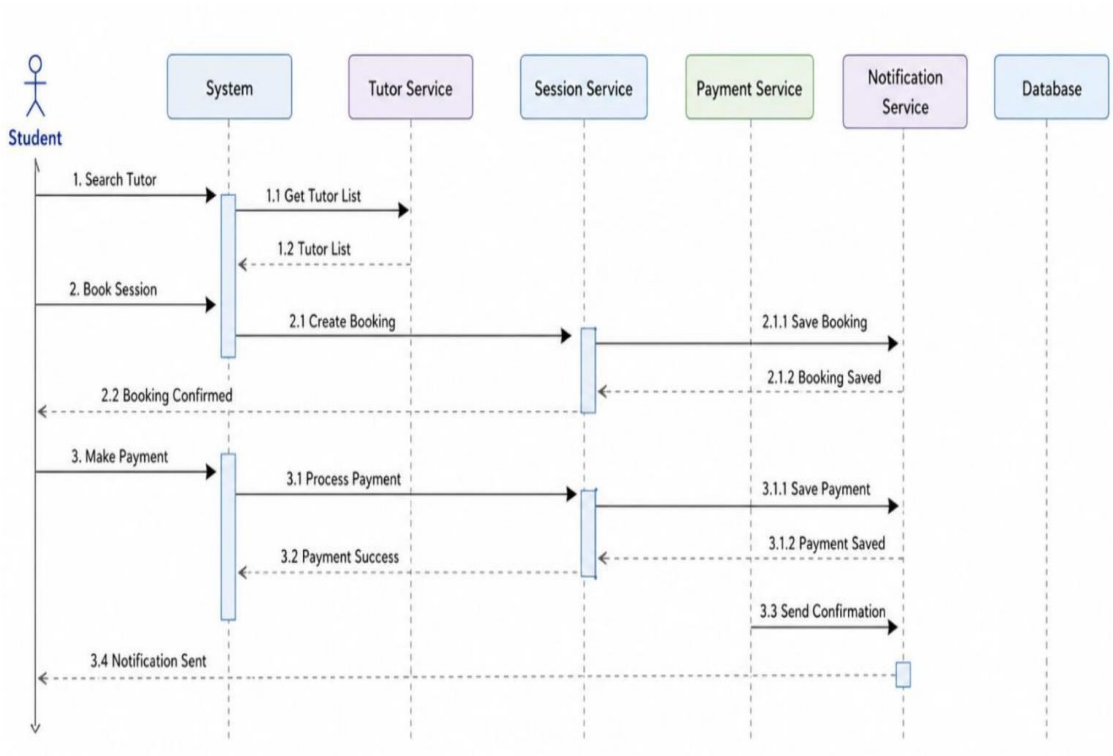


Figure 4.3.3: Sequence Diagram - Student Login & Post

4.4 Entity Relationship (ER) Diagram

The ER Diagram is used to represent the database structure of the system. It shows entities, attributes, and relationships between them.

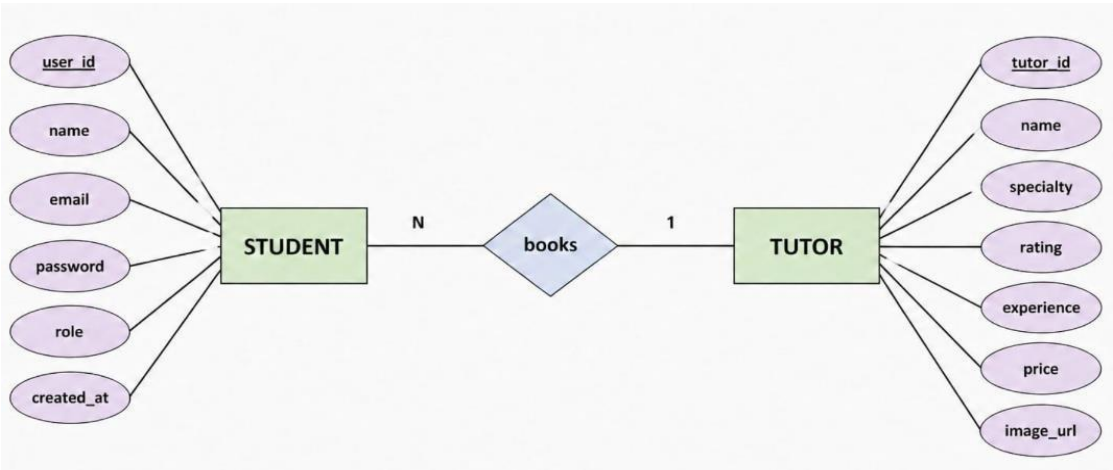


Figure 4.4.1: Entity Relationship (ER) Diagram

Main Entities

1. **User**
   Attributes: User ID, Name, Email, Password
2. **Quiz**
   Attributes: Quiz ID, Title, Questions
3. **Result**
   Attributes: Result ID, Score, Date
4. **Subscription**
   Attributes: Plan ID, Type, Validity

Relationships

- A **User** can attempt multiple quizzes
- A **Quiz** can have multiple results
- A **User** can have one subscription

The ER diagram helps in designing the database efficiently and ensures that data is stored in a structured way.

Additional Design Considerations

To make the system more efficient, certain design principles are followed:

- **Modular Design**: Each feature is developed as a separate module
- **Scalability**: The system can handle increasing users
- **Security**: Data is protected using secure methods
- **Performance Optimization**: Fast response time is ensured.

Flow chart:

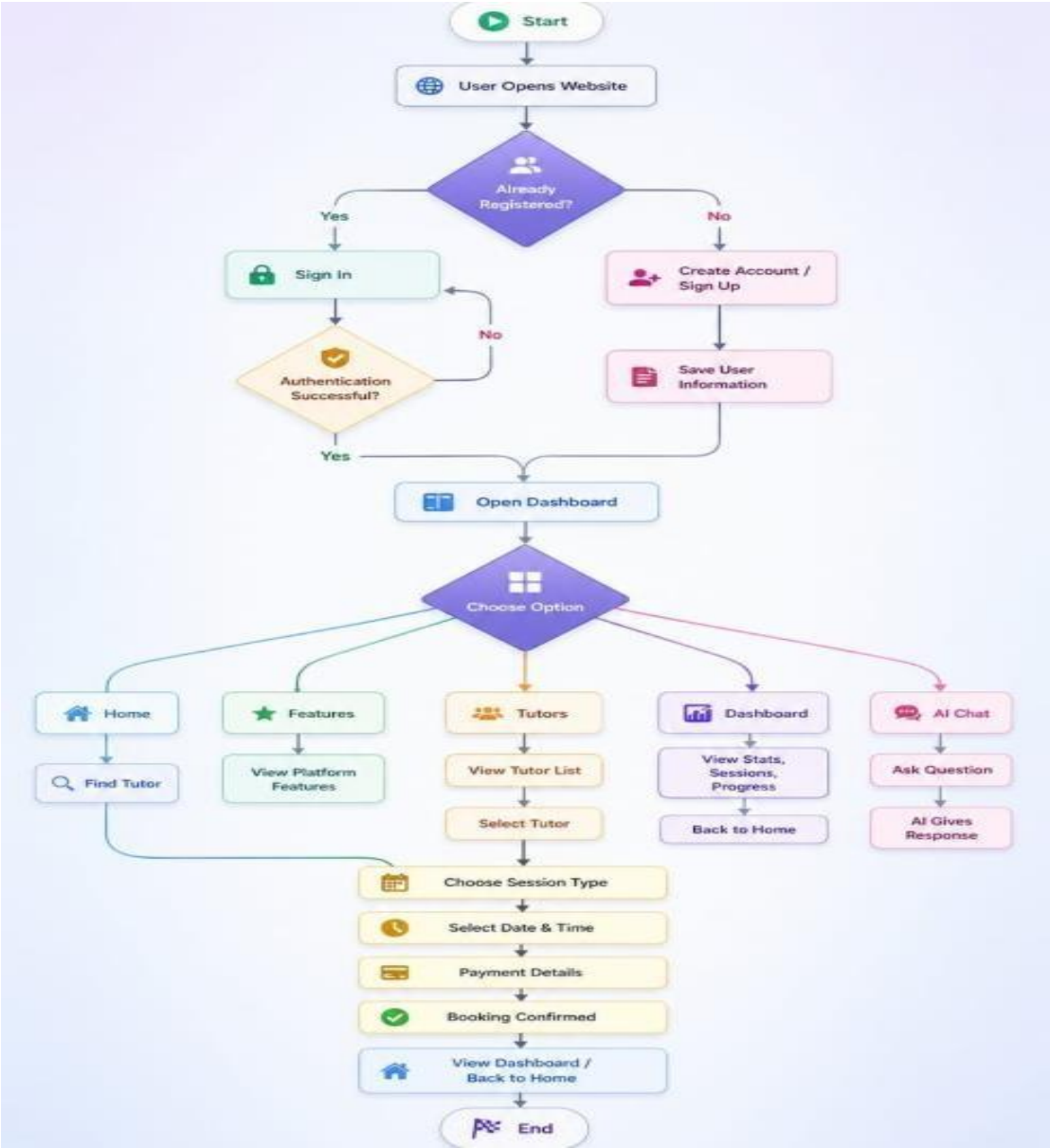


Figure 4.4.2: Flow chart

CHAPTER 5
Implementation & AI Features

5.1 Frontend Implementation

The frontend of the ProTutor system is responsible for providing the user interface through which users interact with the platform. It is developed using **React.js**, which is a popular JavaScript library for building modern and dynamic web applications. React allows the application to be fast, responsive, and easy to manage.

The user interface is designed using **Tailwind CSS**, which helps in creating a clean and responsive layout. The design ensures that the platform works smoothly on different screen sizes such as desktops, laptops, and tablets.

The frontend is divided into different components and pages. Each component is responsible for a specific function, such as login, dashboard, quiz interface, or AI chat. This modular approach makes the system easy to maintain and update.

Some important frontend pages include:

- **Login and Registration Page**
  This page allows users to create an account or log in using their credentials. It also supports social login such as Google.

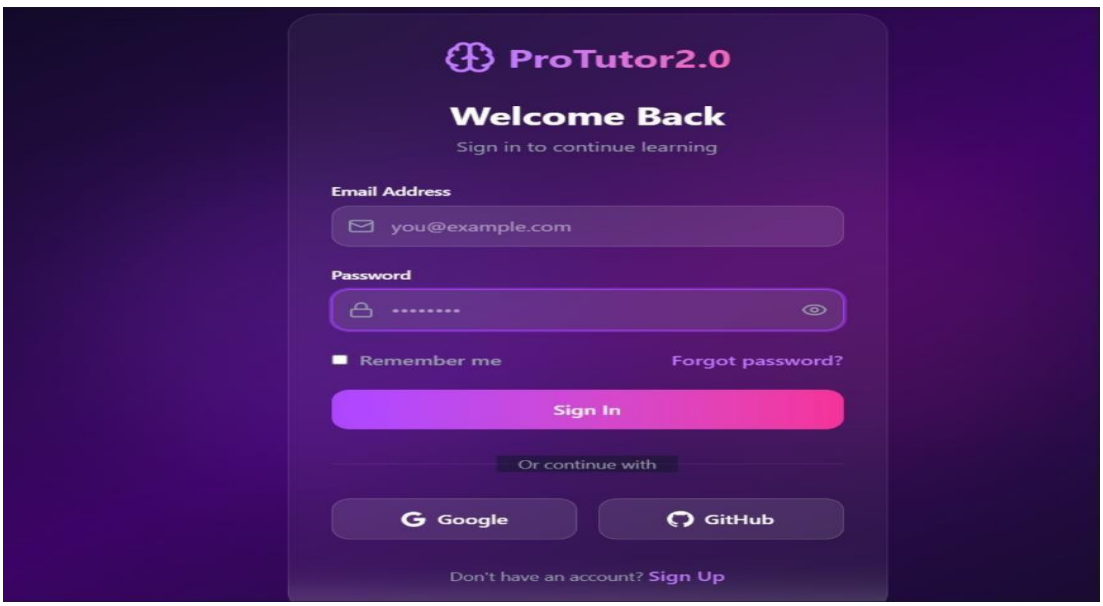


Figure 5.1: User Login Page (Screenshot)

- **Dashboard Page**

  After logging in, users are redirected to the dashboard. This page provides access to all features such as AI tools, quizzes, analytics, and subscription options.

Figure 5.1.2: User Dashboard (Screenshot)

- **AI Chat Interface**

  This interface allows users to ask questions and receive answers from the AI system. It is designed to be simple and interactive.

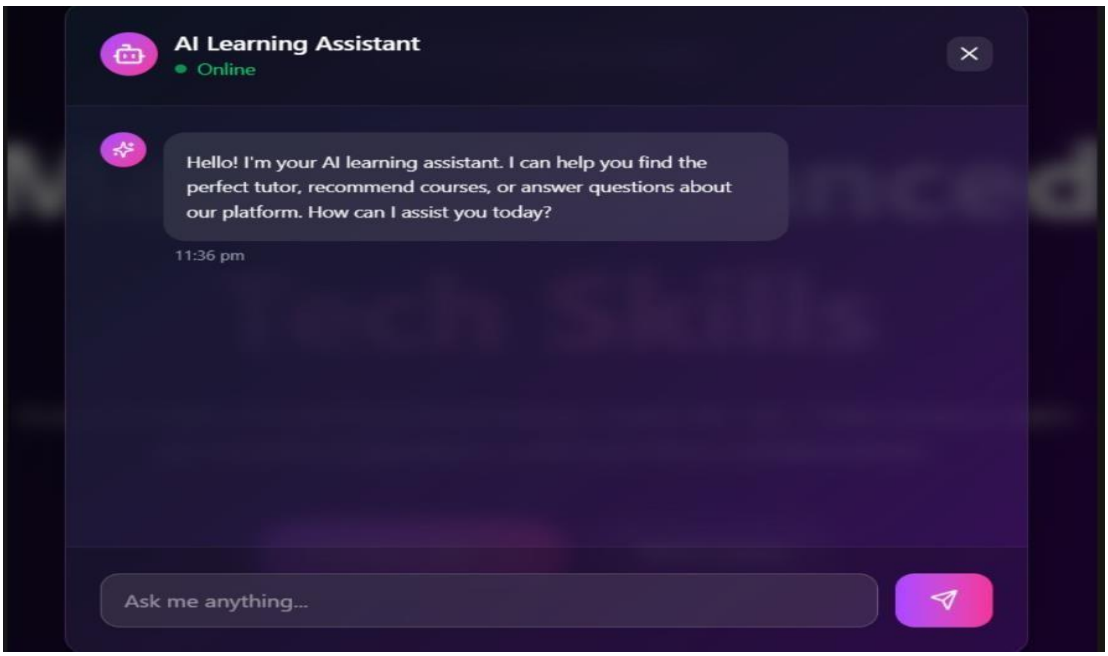


Figure 5.1.3: AI Integration (Screenshot)

- **Quiz Interface**
  Users can attempt quizzes, view questions, and submit answers through this page. Results are displayed after completion.
- **Subscription Page**
  This page shows different plans available for users and allows them to upgrade their accounts.

The frontend communicates with the backend using API calls. When a user performs an action, such as asking a question or starting a quiz, the frontend sends a request to the backend and displays the response.

Overall, the frontend provides a smooth and user-friendly experience, making it easy for students to use the platform.

5.2 Backend Implementation

The backend of the ProTutor system is responsible for handling all the logic and processing. It is developed using **Node.js** and **Express.js**, which are widely used for building server-side applications.

The backend acts as a bridge between the frontend and the database. It receives requests from the frontend, processes them, and sends back the appropriate response.

Some key functionalities of the backend include:

- **User Authentication**
  The backend verifies user credentials during login and registration. It uses secure methods such as JWT (JSON Web Token) to manage user sessions.
- **API Development**
  The backend provides various APIs for different features such as AI queries, quiz generation, user data, and analytics. These APIs allow the frontend to interact with the system.
- **Business Logic Handling**
  The backend contains logic for features like quiz evaluation, performance tracking, and subscription management.

- **AI Integration**
  The backend connects with AI services to process user queries and generate responses. It sends the user's input to the AI system and returns the result.
- **Error Handling**
  The backend ensures that errors are handled properly and meaningful messages are sent to the user.

The backend is designed in a modular way, where different routes and controllers are used for different functionalities. This makes the system organized and easy to maintain.

5.3 Database Implementation

The database is an important part of the ProTutor system as it stores all user data and system information. The system uses **PostgreSQL**, which is a powerful and reliable relational database management system.

The database stores different types of data such as user details, quiz information, results, and subscription plans. The data is organized into tables, and relationships are maintained between these tables.

Some main tables in the database include:

- **Users Table**
  Stores user information such as name, email, password, and account details.
- **Quiz Table**
  Stores quiz-related data such as questions, options, and correct answers.
- **Results Table**
  Stores user performance data such as scores and attempt history.
- **Subscription Table**
  Stores details about user plans and validity.

The database is designed using normalization techniques to reduce data redundancy and improve efficiency. Relationships between tables are defined using keys such as primary keys and foreign keys.

The backend interacts with the database to store and retrieve data whenever required. For example, when a user completes a quiz, the result is stored in the database, and when the user checks performance, the data is retrieved and displayed.

The database also ensures data security by restricting unauthorized access and using proper validation techniques.

Table 5.3: Database schema

S.No	Table Name	Field Name	Data Type	Description
1	Student	student_id	INT	Unique ID of student
2	Student	name	VARCHAR(100)	Student name
3	Student	email	VARCHAR(100)	Student email
4	Student	password	VARCHAR(100)	Student login password
5	Tutor	tutor_id	INT	Unique ID of tutor
6	Tutor	name	VARCHAR(100)	Tutor name
7	Tutor	subject	VARCHAR(100)	Tutor subject/specialization
8	Tutor	experience	VARCHAR(50)	Tutor teaching experience
9	Tutor	rating	FLOAT	Tutor rating
10	Session	session_id	INT	Unique ID of booked session
11	Session	student_id	INT	Reference to student
12	Session	tutor_id	INT	Reference to tutor
13	Session	session_date	DATE	Date of session
14	Session	session_time	TIME	Time of session
15	Session	status	VARCHAR(50)	Booking status

5.4 AI Features

Artificial Intelligence is one of the most important parts of the ProTutor system. It makes the platform smart and helps students learn more effectively.

AI-Based Doubt Solving

This feature allows students to ask questions and get instant answers. The user enters a query, and the system processes it using AI. The AI generates a response based on the input and returns it to the user. This helps students understand concepts quickly without searching multiple sources.

Quiz Generation

The AI system can automatically generate quizzes based on selected topics. It creates questions, options, and answers. This helps students practice different topics without manually searching for questions.

Content Generation

ProTutor can generate study materials such as summaries, notes, and explanations. This feature helps students in revision and saves time. Instead of writing notes manually, students can use AI-generated content.

Performance Analytics

The system analyzes user activity and performance data. It provides insights such as scores, strengths, and weak areas. This helps students focus on areas that need improvement.

Working Process of AI

1. User inputs a query or selects a feature
2. Frontend sends the request to backend
3. Backend processes the request and sends it to AI service
4. AI generates response
5. Backend sends response to frontend

6. User sees the result

This process ensures fast and accurate responses.

Additional Implementation Features

Authentication System

The system uses secure login methods. Users can log in using email or Google. JWT tokens are used to maintain user sessions securely.

Subscription System

ProTutor follows a subscription-based model. Users can access limited features for free, while premium users can unlock advanced features. The system manages user plans and access control.

User Experience

The platform is designed to be simple and easy to use. Clear navigation, fast loading, and responsive design improve the overall user experience.

Advantages of Implementation

- Fast and responsive system
- Secure user data
- Scalable architecture
- Easy to maintain
- Smart AI features

CHAPTER 6
Software Testing

6.1 Unit Testing

Unit testing is the process of testing individual components or modules of the system separately. In the ProTutor project, each feature such as login, quiz generation, AI response, and dashboard is tested individually to ensure that it works correctly.

The main objective of unit testing is to identify errors at an early stage. Each module is tested with different inputs to check whether the output is correct or not. For example, the login module is tested by entering valid and invalid credentials to verify its behavior.

In the ProTutor system, the following components are tested under unit testing:

- **Login and Registration Module**
  Checked whether users can successfully register and log in with correct details. Invalid inputs are also tested.
- **AI Doubt Solving Module**
  Tested by giving different types of questions to ensure that the AI provides correct and relevant answers.
- **Quiz Module**
  Verified whether quizzes are generated properly and results are calculated correctly.
- **Dashboard Module**
  Tested to check whether all user data and features are displayed properly.

Unit testing helps in improving code quality and ensures that each part of the system functions correctly before integrating it with other modules.

Table 6.1 Unit Tast Cases

Test Case ID	Module	Test Case Description	Input	Expected Output	Status
UT-01	Registration	Check user registration with valid details	Name, Email, Password	Account created successfully	Pass
UT-02	Login	Check login with valid email and password	Valid Email, Password	User login successful	Pass
UT-03	Login	Check login with invalid password	Email, Wrong Password	Error message displayed	Pass
UT-04	Tutor Search	Check tutor search by subject	Subject name	Relevant tutor list displayed	Pass
UT-05	Tutor Details	Check tutor profile view	Selected tutor	Tutor details displayed	Pass
UT-06	Session Booking	Check booking with tutor, date, and time	Tutor, Date, Time	Booking details saved	Pass
UT-07	Payment	Check payment details submission	Card details	Payment processed successfully	Pass
UT-08	Dashboard	Check dashboard data display	Logged-in user	Sessions and progress displayed	Pass

UT-09	AI Chat	Check chatbot response	User query	Chatbot reply displayed	Pass

6.2 Integration Testing

Integration testing is the process of testing how different modules of the system work together. After individual components are tested, they are combined and tested as a complete system.

Table 6.2: Integration Testing

IT ID	Modules Integrated	Test Input	Expected Output
IT-01	Registration + Database	Valid student details	User account saved successfully
IT-02	Login + Dashboard	Valid email and password	User redirected to dashboard
IT-03	Tutor Search + Tutor Details	Subject: Maths	Tutor list and details displayed
IT-04	Booking + Tutor Module	Selected tutor	Tutor added to booking process
IT-05	Booking + Date & Time	Selected date and time	Session slot selected successfully
IT-06	Booking + Payment	Valid booking and payment details	Payment processed successfully
IT-07	Payment + Database	Successful payment	Booking data stored successfully
IT-08	Database + Dashboard	Logged-in user	Sessions and progress displayed

IT-09	AI Chat + User Interface	User query	Chatbot response shown on screen
IT-10	Complete System Flow	Login → Search Tutor → Book Session → Pay	Booking confirmed and dashboard updated

In ProTutor, integration testing ensures that the frontend, backend, and database are properly connected. It verifies that data flows correctly between different parts of the system.

Some important integration testing scenarios include:

- **Frontend and Backend Integration**
  When a user performs an action on the frontend, such as logging in or starting a quiz, the request is sent to the backend. The backend processes the request and sends the response back to the frontend. This interaction is tested to ensure smooth communication.
- **Backend and Database Integration**
  The backend interacts with the database to store and retrieve data. For example, when a user completes a quiz, the result is stored in the database. This process is tested to ensure accuracy.
- **AI Integration**
  The system is tested to ensure that AI responses are correctly generated and displayed to the user. The communication between backend and AI service is verified.

Integration testing helps in identifying issues that may occur when different modules interact with each other. It ensures that the system works as a whole without errors.

6.3 Test Cases

Test cases are used to verify whether the system is working as expected. Each test case includes input, expected output, and actual result. Below are some important test cases for the ProTutor system:

These test cases ensure that all major functionalities of the system are working properly.

Testing Methods Used

Different testing methods are used to ensure system quality:

- **Black Box Testing**
  Tests system functionality without looking at internal code.
- **White Box Testing**
  Tests internal logic and code structure.
- **Manual Testing**
  Testing performed by manually using the system.

These methods help in identifying errors and improving system performance.

Advantages of Testing

- Improves system reliability
- Identifies bugs and errors
- Ensures proper functionality
- Enhances user experience
- Increases system performance

Conclusion of Testing

Testing is an important part of system development. In the ProTutor project, proper testing ensures that all features are working correctly and efficiently. Both unit testing and integration testing are performed to verify system functionality. Test cases help in validating expected results.

Overall, the system is tested successfully and is ready for deployment.

Table 6.3 Test Cases

TCID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-01	User Registration	Open register page, enter details, click Sign Up	Name, Email, Password	New account should be created	Account created successfully	Pass
TC-02	User Login	Open login page, enter valid credentials	Valid login credentials	User should login and redirect to dashboard	User redirected to dashboard	Pass
TC-03	Invalid Login	Enter wrong password and click Login	Wrong password	System should show error message	Error message displayed	Pass
TC-04	Tutor Search	Enter subject in search bar	Maths / Java	Relevant tutors should be displayed	Tutor list displayed correctly	Pass
TC-05	View Tutor Details	Select a tutor profile	Tutor name / subject	Tutor details should be visible	Tutor details displayed	Pass
TC-06	Book Session	Select tutor, date, and time	Tutor + Date + Time	Booking should move to payment step	Payment step opened	Pass

TC-07	Payment Success	Enter valid payment details and submit	Valid card/payment details	Payment should be successful	Payment successful	Pass
TC-08	Payment Failure	Enter invalid payment details	Invalid card/payment details	Payment failed message should appear	Payment failed message displayed	Pass
TC-09	Booking Confirmation	Complete successful payment	Payment success	Booking confirmation should be shown	Booking confirmed	Pass
TC-10	Dashboard View	Login and open dashboard	Logged-in user	Sessions, progress and statistics displayed	Dashboard data displayed	Pass
TC-11	AI Chat Support	Enter a question in chatbot	User query	Chatbot should return a response	Chatbot response displayed	Pass
TC-12	Logout	Click logout button	Logged-in user	Session should end and return to login page	User logged out successfully	Pass

CHAPTER 7
Conclusion & Summary

7.1 Conclusion

The ProTutor project successfully demonstrates the design and development of a full-stack AI-powered learning platform aimed at improving the academic experience of students. The main goal of this project was to create a centralized system where students can perform all their learning activities in one place. This objective has been achieved by integrating various features such as AI-based doubt solving, quiz generation, content creation, performance tracking, and resume guidance.

In today's digital learning environment, students often depend on multiple platforms for studying, practicing, and clearing doubts. This creates confusion and wastes time. ProTutor solves this problem by providing a single, user-friendly platform that combines all essential tools. This makes the learning process more organized, efficient, and effective.

The system is developed using the PERN stack, which includes PostgreSQL, Express.js, React.js, and Node.js. This technology stack ensures that the application is fast, scalable, and reliable. The frontend provides an interactive and responsive interface, while the backend handles logic, data processing, and communication with the database. The database stores user data securely and efficiently.

One of the most important aspects of this project is the integration of Artificial Intelligence. AI helps in providing instant answers to student queries, generating quizzes, and creating study content. This makes the platform more intelligent and useful for students. It also reduces the time required to search for information and improves learning efficiency.

The system also includes a performance tracking feature that helps students analyze their progress. By identifying strengths and weaknesses, students can focus on improving their weak areas. This leads to better academic performance.

In addition, the platform follows a SaaS-based model with free and premium features. This allows users to access basic features for free while providing advanced features through subscriptions. This model makes the platform accessible and sustainable.

Testing has been performed to ensure that all features work correctly. Unit testing and integration testing were used to verify system functionality. The results show that the system is reliable and ready for use.

Overall, the ProTutor project is a successful implementation of a modern learning platform. It not only provides a practical solution for students but also demonstrates real-world full-stack development skills. The project covers all aspects of software development, including frontend, backend, database, security, and AI integration.

7.2 Project Summary

ProTutor is a full-stack AI-powered learning platform designed to help students improve their academic performance through a single, centralized system. The main aim of the project is to solve the problem of scattered learning resources by combining multiple features such as AI-based doubt solving, quiz generation, content creation, performance tracking, and resume guidance into one platform.

The system is developed using the PERN stack (PostgreSQL, Express.js, React.js, Node.js), which ensures good performance, scalability, and reliability. The frontend provides a user-friendly interface, while the backend handles system logic and communication with the database. Artificial Intelligence plays a key role by providing instant answers, generating quizzes, and assisting in learning.

ProTutor follows a SaaS-based model where users can access basic features for free and advanced features through subscription plans. The platform is designed to be simple, efficient, and accessible for students.

Overall, the project demonstrates a complete full-stack development process and provides a practical solution for modern digital learning by making education more organized, interactive, and effective.

APPENDICES

APPENDIX 1: ABBREVIATIONS / NOMENCLATURE

This section lists the important abbreviations and technical terms used in the ProTutor project report. These abbreviations help in understanding the report more easily.

List of Abbreviations

S. No	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	UI	User Interface
3	UX	User Experience
4	API	Application Programming Interface
5	DB	Database
6	SQL	Structured Query Language
7	JWT	JSON Web Token
8	HTTP	HyperText Transfer Protocol
9	HTTPS	HyperText Transfer Protocol Secure
10	SaaS	Software as a Service
11	PERN	PostgreSQL, Express.js, React.js, Node.js
12	DFD	Data Flow Diagram
13	UML	Unified Modeling Language
14	ER Diagram	Entity Relationship Diagram
15	IDE	Integrated Development Environment

S. No	Abbreviation	Full Form
16	OS	Operating System
17	RAM	Random Access Memory
18	CPU	Central Processing Unit
19	CSS	Cascading Style Sheets
20	HTML	HyperText Markup Language
21	JS	JavaScript
22	REST	Representational State Transfer
23	CRUD	Create, Read, Update, Delete
24	JSON	JavaScript Object Notation

Conclusion

The above abbreviations are used throughout the project report to simplify technical terms and improve readability. Understanding these terms will help the reader easily follow the concepts and implementation of the ProTutor system.

APPENDIX 2: **References**

The following references were used during the development of the ProTutor project. These sources include books, websites, documentation, and research materials related to full-stack development, databases, and artificial intelligence.

Books

1. Elmasri, R., & Navathe, S. B.
   Fundamentals of Database Systems
   Pearson Education.

2. Pressman, R. S.
   Software Engineering: A Practitioner's Approach
   McGraw-Hill.

3. Sommerville, I.
   Software Engineering
   Pearson Education.

4. Duckett, J.
   JavaScript and JQuery: Interactive Front-End Web Development
   Wiley Publications.

5. Freeman, E.
   Head First Design Patterns
   O'Reilly Media.

Websites and Online Resources

6. React.js Official Documentation
   https://react.dev

7. Node.js Official Documentation
   https://nodejs.org

8. Express.js Official Documentation
   https://expressjs.com

9. PostgreSQL Official Documentation
   https://www.postgresql.org

10. Tailwind CSS Documentation
    https://tailwindcss.com

11. MDN Web Docs (HTML, CSS, JavaScript)
    https://developer.mozilla.org

12. W3Schools Tutorials
    https://www.w3schools.com

13. Stack Overflow
https://stackoverflow.com

AI and Technology Resources

14. OpenAI Documentation (AI APIs and usage)
https://platform.openai.com

15. GeeksforGeeks
https://www.geeksforgeeks.org

16. TutorialsPoint
https://www.tutorialspoint.com

17. FreeCodeCamp
https://www.freecodecamp.org

Research and Learning Materials

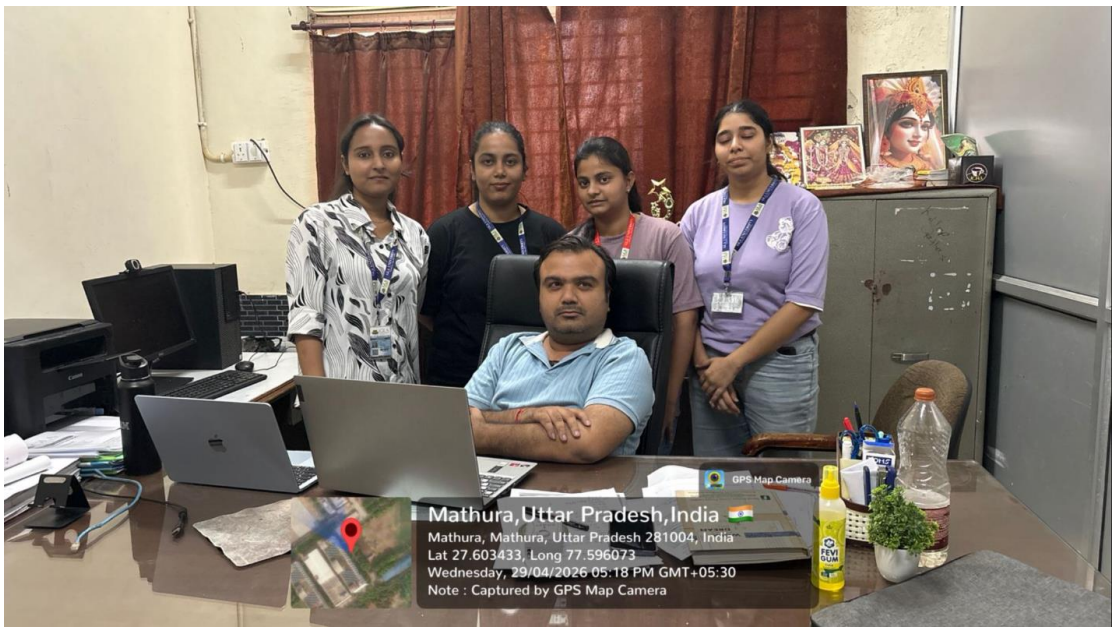
18. Various online articles and blogs related to full-stack development and AI integration

19. YouTube educational channels for web development tutorials

20. College study materials and lecture notes

Conclusion of References

The above references provided valuable knowledge and guidance during the development of the ProTutor project. These sources helped in understanding concepts related to frontend development, backend programming, database management, and artificial intelligence. The use of official documentation ensured accurate implementation, while books and tutorials provided theoretical and practical knowledge. Overall, these references played an important role in completing the project successfully.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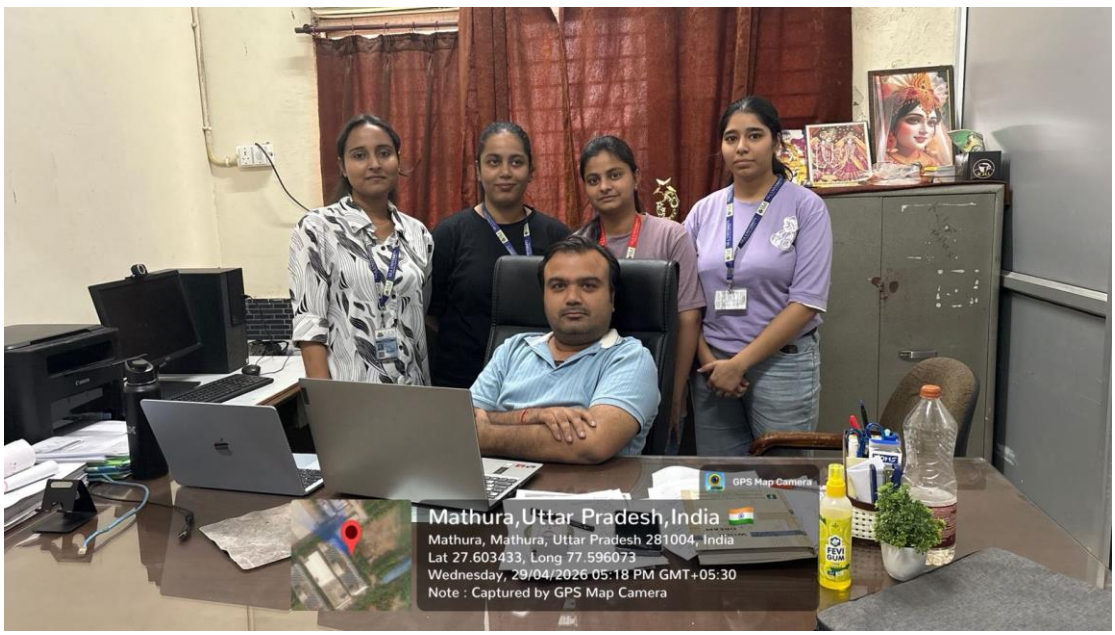
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